**CURRICULUM VITAE**
**SEBASTIANO NICOLUSSI GOLO**

Foto scattata da Tuomas Orponen


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| Name: | Sebastiano, Nicolussi Golo |
| Date of birth: | March 17, 1986 |
| Citizenship: | Italian |
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## Education

Abilitazione Scientifica Nazionale Italiana: 28 February 2025
Settore Concorsuale: 01/A3, Fascia: 2, Quadrimestre n. 3.
Domanda n.126884.

Doctor of Philosophy in Mathematics: 25 October 2017
Double degree based on a cotutelle agreement:
University of Jyväskylä, Finland
Advisor: *Enrico Le Donne*
University of Trento, Italy
Advisor: *Francesco Serra Cassano*
Thesis: *Topics in the geometry of non-Riemannian Lie groups*
Grade: *Pass with distinction*

Master degree in mathematics: March 2012
Università degli Studi di Trento, Italia
Supervisor: *Francesco Serra Cassano*
Thesis: *The tangent space in subRiemannian geometry*
Grade: *110/110 cum laude*

Bachelor degree in mathematics: July 2009
Università degli Studi di Trento, Italia
Thesis: *Symplectic geometry*
Supervisor: *Roberto Pignatelli*
Grade: *108/110*

## Employments and Fellowships

Senior Researcher (Oberassistent) at the University of Fribourg
March 2025 – August 2025

Senior Researcher (Senior Forscher) at the University of Fribourg
May 2024 – February 2025

A pause of one year for religious research
September 2023 – April 2024

Postdoctoral Researcher (Tutkijatohtori) at the University of Jyväskylä
September 2022 – August 2023
Mentor: Enrico Le Donne

Postdoctoral Researcher (Tutkijatohtori) at the University of Jyväskylä
October 2020 – August 2022
Mentor: Katrin Fässler

Postdoctoral Researcher at the University of Padova
December 2018 – June 2020
Mentor: Davide Vittone

Postdoctoral Researcher at the University of Birmingham
December 2017 – November 2018
Mentor: Alessio Martini

Visiting Student at the University of Granada
November 2017
Mentor: Manuel Ritoré

Postgraduate Student within the project Marie Curie ITN MAnET
January 2015 – October 2017

Postgraduate Student at the University of Jyväskylä, Finland
October 2012 – December 2014

## Grants and Awards

Seal of Excellence from Marie Skłodowska-Curie actions:
The project proposal 101025614, *MICA: Minimal Submanifolds in Carnot Groups*, Submitted under the Horizon 2020 Marie Skłodowska-Curie actions call H2020-MSCA-IF-2020 of 9 September 2020 by Sebastiano Nicolussi Golo and Jyvaskylan Yliopisto was scored as a high-quality project proposal in a highly competitive evaluation process, with a total score of 92.20%. This proposal is recommended for funding by other sources, since Horizon 2020 resources available for this specific Call were already allocated following a competitive ranking.

Seal of Excellence from Marie Skłodowska-Curie actions:
The project proposal 888619, *MICA: Minimal Submanifolds in Carnot Groups*, Submitted under the Horizon 2020 Marie Skłodowska-Curie actions call H2020-MSCA-IF-2019 of 11 September 2019 by Sebastiano Nicolussi Golo and Universidad de Granada was scored as a high-quality project proposal in a highly competitive evaluation process, with a total score of 87.40%. This proposal is recommended for funding by other sources, since Horizon 2020 resources available for this specific Call were already allocated following a competitive ranking.

INdAM–GNAMPA Project 2019 "Rectifiability in Carnot Groups"
(as Principal Investigator)
March 2019 – February 2020

## Publications

[1] F. Bigolin and S. Nicolussi Golo. A historical account on characterizations of $C^1$-manifolds in Euclidean spaces by tangent cones. *J. Math. Anal. Appl.*, 412(1):63–76, 2014.

[2] E. Le Donne, S. Nicolussi Golo, and A. Sambusetti. Asymptotic behavior of the Riemannian Heisenberg group and its horoboundary. *Ann. Mat. Pura Appl. (4)*, 196(4):1251–1272, 2017.

[3] S. Nicolussi Golo. Topics in the geometry of non-Riemannian Lie groups. PhD thesis, *Report*, Department of Mathematics and Statistics, University of Jyväskylä, 163 (2017).

[4] E. Le Donne and S. Nicolussi Golo. Regularity properties of spheres in homogeneous groups. *Trans. Amer. Math. Soc.*, 370(3):2057–2084, 2018.

[5] S. Golo. Some remarks on contact variations in the first Heisenberg group. *Ann. Acad. Sci. Fenn. Math.*, 43(1):311–335, 2018.

[6] S. Golo and S. Rigot. The Besicovitch covering property in the Heisenberg group revisited. *J. Geom. Anal.*, 29(4):3345–3383, 2019.

[7] S. Nicolussi and F.Serra Cassano. The Bernstein problem for Lipschitz intrinsic graphs in the Heisenberg group. *Calc. Var. Partial Differential Equations*, 58(4):Art. 141, 28, 2019.

[8] E. Le Donne and S. Nicolussi Golo. Metric Lie groups admitting dilations. *Ark. Mat.*, 59(1) (2021).

[9] S. Nicolussi Golo and M. Ritoré. Area-minimizing cones in the Heisenberg group H. *Ann. Fenn. Math.*, 46(2):945–956, 2021.

[10] N. Fisher and S. Nicolussi Golo. Sub-Finsler horofunction boundaries of the Heisenberg group. *Anal. Geom. Metr. Spaces*, 9(1):19–52, 2021.

[11] A. Julia and S. Nicolussi Golo. Intrinsic rectifiability via flat cones in the Heisenberg group. *Ann. Sc. Norm. Super. Pisa Cl. Sci.*, Vol. XXIII, Issue 2 (2022).

[12] A. Julia, S. Nicolussi Golo, and D. Vittone. Nowhere differentiable intrinsic Lipschitz graphs. *Bull. Lond. Math. Soc.*, 53(6) (2021).

[13] A. Martini, D. Müller, and S. Nicolussi Golo. Spectral multipliers and wave equation for sub-Laplacians: lower bounds of Euclidean type. *J. Eur. Math. Soc. (JEMS)* (2021).

[14] A. Julia, S. Nicolussi Golo, and D. Vittone. Area of intrinsic graphs and coarea formula in Carnot groups. *Math. Z.* 301, 1369–1406 (2022).

[15] E. Le Donne, V. Kivioja, and S. Nicolussi Golo. Metric equivalences of Heintze groups and applications to classifications in low dimension. *Illinois J. Math.* 66 (1) 91 - 121, April 2022.

[16] A. Julia, S. Nicolussi Golo, and D. Vittone. Lipschitz functions on submanifolds in Heisenberg groups. *International Mathematics Research Notices, Volume 2023, Issue 9, May 2023, Pages 7399–7422.*

[17] G. Antonelli, E. Le Donne, and S. Nicolussi Golo. Lipschitz Carnot-Carathéodory structures and their limits. *J. Dyn. Control. Syst. 29, 805–854* (2023).

[18] S. Eriksson-Bique, C. Gartland, E. Le Donne, L. Naples, and S. Nicolussi Golo. Nilpotent groups and biLipschitz embeddings into L1. *International Mathematics Research Notices, Volume 2023, Issue 12, June 2023, Pages 10759–10797.*

[19] S. Nicolussi Golo, and B. Warhurst. Jet spaces over Carnot groups. *Rev. Mat. Iberoam. 39 (2023), no. 6, pp. 2289–2330.*

[20] L. Ambrosio, S. Nicolussi Golo, and F. Serra Cassano. Optimal $C^\infty$-approximation of functions with exponentially or sub-exponentially integrable derivative. *Calc. Var. Partial Differ. Equ. 62 21* (2023).

[21] L. Ambrosio, S. Nicolussi Golo, and F. Serra Cassano. Classical flows of vector fields with exponential or sub-exponential summability. *J. Differ. Equations 372 458–504* (2023).

[22] A. Ottazzi, and S. Nicolussi Golo. Left-invariant distributions diffeomorphic to flat distributions. *Geom. Dedicata 218, 56* (2024).

[23] S. Nicolussi Golo and Y. Zhang. Curvature Exponent and Geodesic Dimension on Sard-regular Carnot groups. *Anal. and Geom. in Metr. Spaces, vol. 12, no. 1, pp. 20240004* (2024).

[24] M. G. Cowling, V. Kivioja, E. Le Donne, S. Nicolussi Golo, and A. Ottazzi. From homogeneous metric spaces to Lie groups. *C. R. Math. Acad. Sci. Paris 362 (2024), 943–1014.*

[25] M. Di Marco, A. Julia, S. Nicolussi Golo, and D. Vittone. Submanifolds with boundary and Stokes' Theorem in Heisenberg groups. *Trans. Amer. Math. Soc. 378 (2025), 4955-4990.*

[26] C. Y. Guo, S. Nicolussi Golo, M. Williams, and Y. Xuan. Quasiregular mappings between subRiemannian manifolds. *J. Lond. Math. Soc. (2) 112 (2025), no. 2, Paper No. e70254.*

[27*] S. Nicolussi Golo and B. Warhurst. Moduli of Sub-Laplacians on the second Heisenberg group. *arXiv e-prints.*

[28*] A. Kijowski, S. Nicolussi Golo, and B. Warhurst. Equivalence of sub-Laplacian on Polarized groups. *arXiv e-prints.*

[29*] E. Le Donne, L. Nalon, S. Nicolussi Golo, and S. Y. Ryoo. Asymptotics of Riemannian Lie groups with nilpotency step 2. *arXiv e-prints.*

## Research stays

- Visiting Student at "Laboratoire J.A. Dieudonné", Nice, France (November–December 2016)
- Visiting Graduate Student at the Department of Mathematics of Tufts University, Medford, USA (January–February 2017)
- Visiting Graduate Student at the Simons Semester "Symmetry and Geometric Structures", Warsaw, Poland (August–October 2017)
- Junior Leader of the Simons Semester at IMPAN in Warsaw (September–October 2019)
- Theoretical Sciences Visiting Program (TSVP) visitor to OIST, *Okinawa Institute of Science and Technology*, in Japan (August–October 2022)

## Activities

- Teaching assistant ("Geometry of surfaces in 3D space"): University of Jyväskylä, Finland (Fall 2012)
- Organizer of the "Kindergarten Seminar" for doctoral students at the University of Jyväskylä, Finland (Spring 2014)
- Organizer of the seminar for doctoral students at Institute Henri Poincaré, Paris (October 2014)
- Co-organizer of the conference "Subriemannian Geometry and Beyond", Jyväskylä, Finland (February 2018)
- Co-organizer of the conference "Harmonic Analysis and PDE Day", Birmingham, UK (September 2018)
- Organizer of seminars "GeA (Geometria e Analisi)" at the Department of Mathematics, University of Padova (Fall 2019)
- Teaching assistant ("Analysis 1"): University of Padova, Faculty of Engineering (Fall 2019)
- Co-organizer of the workshop "Geometric Measure Theory in Padova", Padova, Italy (January 2020)
- Representative of postdocs in the department council of the Department of Mathematics at the University of Padova (January–June 2020)
- Master thesis supervisor; University of Jyväskylä (2021–2022)
- Teacher of the course "Riemannian Geometry"; University of Jyväskylä, (2021–2022)
- Teacher of the Mini-course "BV functions in Carnot groups"; OIST, Japan (August 2022)
- Co-organizer in the conference "Sub-Riemannian Geometry and Beyond, III", Pisa (June 2023)
- Teacher of the course "Thematical Seminar"; University of Fribourg, Switzerland (Fall 2024)
- Teacher of the course "Introduction to PDE"; University of Fribourg, Switzerland (Spring 2025)
- Co-organizer in the conference "Sub-Riemannian Geometry and Beyond, IV", Schwarzsee, Switzerland (June 2025)